

Axis B, tractability

A declarative encoding of the breakdown radius — what it buys, what it costs, and what it cannot claim

	Result
The encoding works	Three encodings (E1, E2, E3), validated against brute-force BFS and local_up on 1,044 exhaustive instances at $n = 4$ and 360 generated instances to $n = 20$. Zero disagreements.
It does not replace local_up	On the SAT side it loses badly: when a failure exists, local_up median 0.0027 s against E1's 0.099 s. This is the session's main negative result.
It wins where robustness is certified	Proving that <i>no</i> failure exists, the ordering reverses: local_up exceeded a 60 s cap on 26 of 49 instances; E1 answered every one, the worst ($n = 16$, $k = 33$) in 8.745 s.
The crossover is not k	It is the radius . The 1.59^k model describes local_up's worst case, and that case is realised precisely on the degenerate instances.
Conjecture 2 survives	E3 assumes nothing and could have refuted it. Across 1,007 certified witness walks it never once beat E1 — and every margin is exactly 0 , a tie everywhere rather than slack.
Degeneracy dominates	64.2% of generated instances have no reachable failure at all. Any tractability claim that averages over these is measuring the easy case.

Session 3. Continues *report_synth_and_search.md* (session 1) and *report_axisb_deep.md* (session 2), neither of which is modified. Branch **experiments/synth_graphs_and_heuristics**; nothing committed to main. In scope per the brief: the SAT/CP encoding, scaling, and stress-testing Conjecture 2. Explicitly parked and untouched: the Anti-Exchange Case B proof, the L/U bounds work, and Axis A.

1. What was built

Three encodings in **src/bkrobust/sat/**, deliberately not collapsed into one, because they assume different things and the differences are the experiment. All three are **upper** bounds on the true radius: none can return a radius that is too small. That one-sidedness is the dangerous direction — an over-estimated radius claims more robustness than is warranted — so each result object carries an **assumes** field that travels with the number.

	searches	assumes	guarantee
E1	retractions of $K(G_0)$	Conjecture 2	$r \leq r_{E1}$
E2	every closed state, distance via the join	Theorem 2 + Lemma R + up-then-down normalisation	$r \leq r_{E2} \leq r_{E1}$
E3	walks of one-orientation covering steps	nothing	$r \leq r_{E3}$

1.1 Two details that are not incidental

E2's join needs no closure, and no extra copy of the variables. The plan budgeted a second copy for the join J . Theorem 2 identifies $G_0 \vee G$ with $K(G_0) \cap K(G)$, and the intersection of closed sets is closed, so $K(J)$ is that intersection outright. With $K(G_0)$ constant the objective collapses to a symmetric difference — one linear constraint over one copy. The saving is real; the assumptions are unchanged, and they are what matters.

E3's soundness runs in one direction only, and it is easy to invert. Consecutive states in an E3 walk are closed and differ by exactly one orientation, and two sets differing by a single element admit nothing strictly between them. So such a pair is a covering pair *unconditionally* — no appeal to Lemma R, to anti-exchange, or to gradedness — and an E3 walk of length k proves $r \leq k$ outright. The converse, that every cover is a one-orientation step, *is* Lemma R, and is neither needed nor assumed.

$r_{E3} < r_{E1}$	refutes Conjecture 2 , with a witness and no assumptions
$r_{E3} = r_{E1}$	is consistent with Conjecture 2 and proves nothing

Agreement is evidence. Only disagreement would be a theorem, and it would be a theorem in the negative direction.

2. Correctness before scale

A wrong encoding does not crash. It returns plausible numbers. So the validation was layered, and each layer found a real bug before the next ran.

Stage 1 — the closure encoding defines the corrected space

Solutions compared against `enumerate_space_correct` on every CPDAG at $n \leq 4$: **133 CPDAGs, 1,588 elements, 0 mismatches**. It did not pass first time. The first version encoded Meek's rules as implications and dropped R3's and R4's undirectedness premises, reproducing only **66 of 133** spaces — strictly too strong. R1 and R2 tolerate the omission, because the alternatives to their consequent are a new v-structure or a cycle, both independently forbidden. R3 and R4 do not. Fixed by encoding every rule as a forbidden firing configuration.

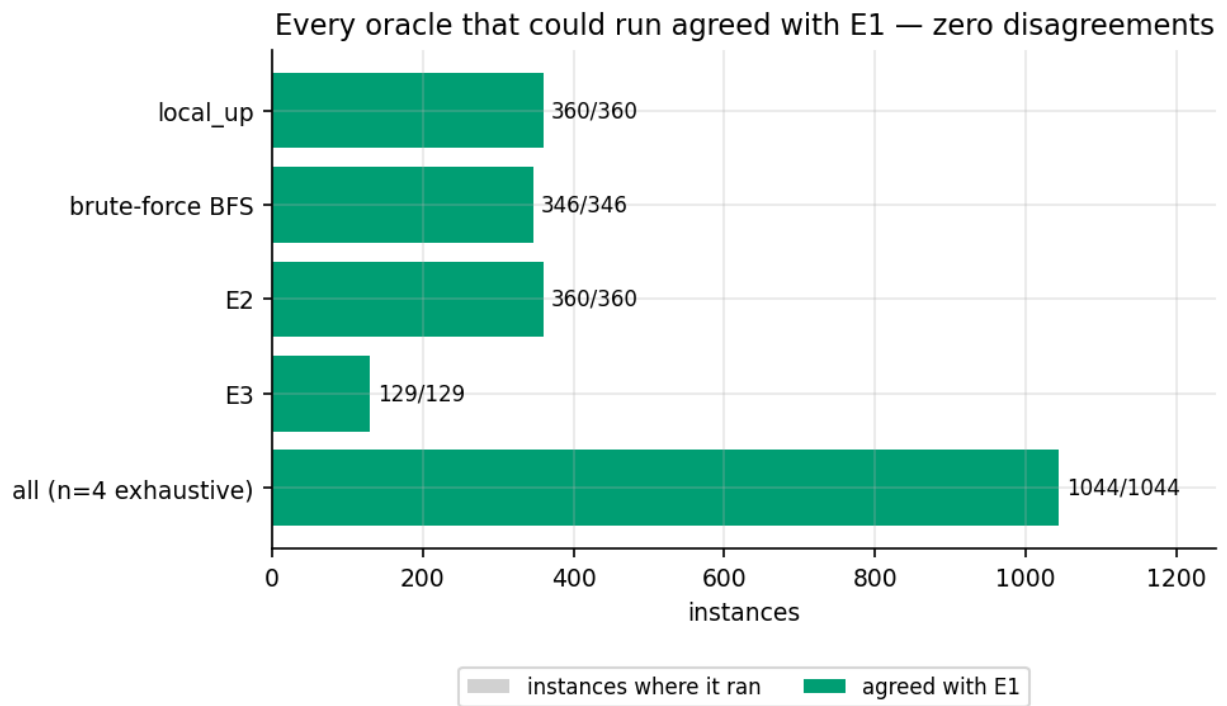
Stage 2 — the failure predicate agrees with the validity oracle

Against the reference oracle: **26,304 cases, 0 disagreements**. It did not pass first time either, and the failure was in the dangerous direction. 144 cases disagreed because the non-collider clause required `into_l v into_r`, whereas a collider needs **both** arrows pointing in. A chain through a member of Z was accepted as a collider, so blocked paths read as open and non-failures read as failures — which makes radii **too small**. A spuriously small E3 radius is exactly what a Conjecture 2 counterexample looks like; had this survived it would have manufactured a false headline.

Stage 3 — radii, differentially, against both existing oracles

comparison	agree	disagree
E1 vs exact BFS	1,044	0
E1 vs local_up	1,044	0
E2 vs exact BFS	1,044	0
E3 vs exact BFS	1,044	0
E1 vs E2	1,044	0

All CPDAGs on 4 nodes, corrected space, Z the optimal adjustment set: 1,044 instances. Of those, 792 have a reachable failure; **all 792 E3 walks were replayed through the ordinary graph code and certified**, and all 792 were monotone — the solver was free to descend and never needed to. E2's optimum had an empty down-leg in all 1,044. On the session-1 worked example all three encodings return the published 3 / 3 / 2, in about 10 ms, without ever building the 48-element space.



Every oracle that could still run agreed with E1.

An honesty fix found during validation

E1's ladder has a natural top — once every orientation is retracted there is nothing left — so exhausting it genuinely proves no failure exists in the up-set. **E3's does not.** Its max_k is a pure budget, so exhausting it proves only radius > max_k. The first version returned UNREACHED there, which would have read as “no failure exists”. It now records exhausted_to and sets exact=False.

Determinism

Radii, witnesses, walks **and solver conflict/branch counters** hash identically across PYTHONHASHSEED 0, 1, 12345, single worker, fixed seed: 4edd011f64bc.... The counters are in the hash deliberately — matching radii alone would not detect a search that explored a different tree and landed on the same answer.

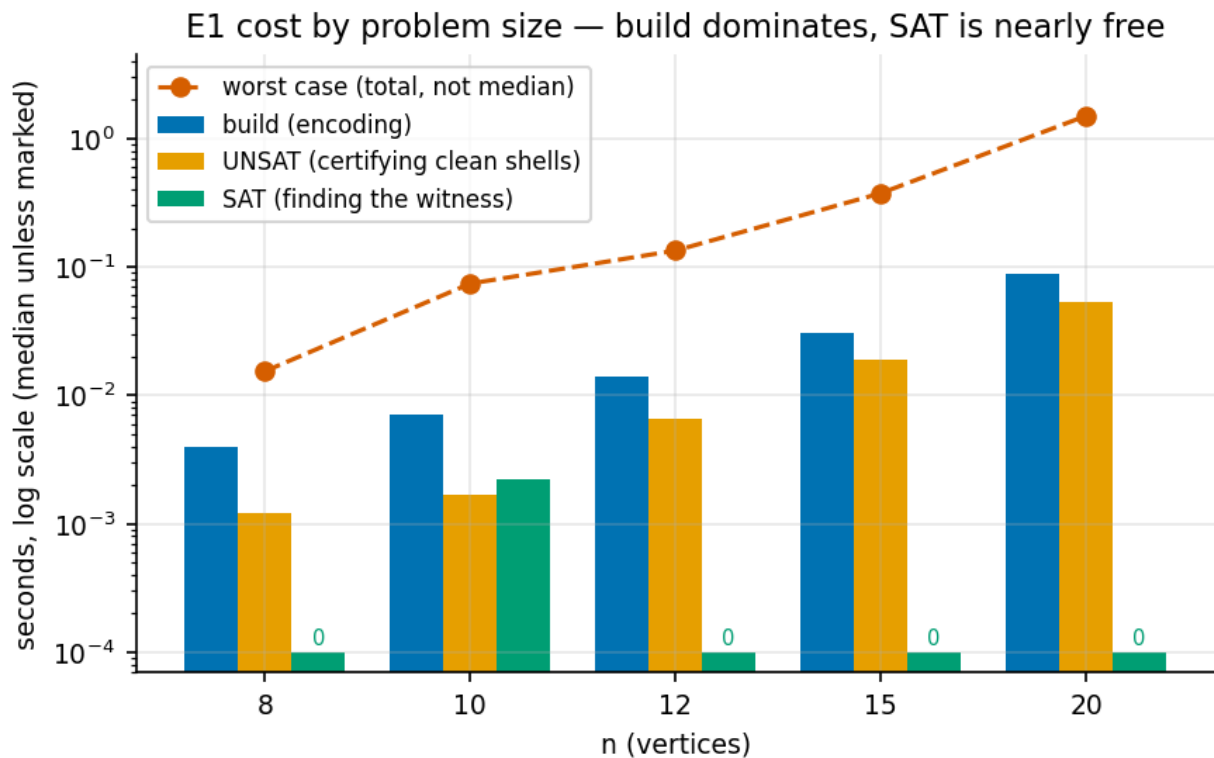
3. Scaling: how far it goes, and where it stops

360 instances across six generators (Erdős-Rényi at three densities, scale-free, block, decoupled-backdoor) at $n = 8, 10, 12, 15, 20$, written incrementally to results/axisb3/scaling.jsonl. Every instance carries its structure, so cost is read against structure rather than against n alone. local_up and BFS were run alongside wherever still feasible, so the comparison is measured, not asserted.

3.1 E1 reaches $n = 20$ without difficulty, and build dominates

n	rows	K med/max	build	UNSAT	SAT	total med	worst
8	72	2 / 7	0.004 s	0.001 s	0.000 s	0.006 s	0.015 s
10	72	3 / 8	0.007 s	0.002 s	0.002 s	0.012 s	0.074 s
12	72	4 / 11	0.014 s	0.007 s	0.000 s	0.025 s	0.134 s
15	72	3 / 9	0.031 s	0.019 s	0.000 s	0.054 s	0.374 s
20	72	4 / 12	0.087 s	0.053 s	0.000 s	0.136 s	1.497 s

The encoding **build**, not the solve, is the bottleneck at this scale, and it grows with n because the Meek clause set ranges over vertex tuples (R3 and R4 over four vertices, so $O(n^4)$ clauses). Median solver conflicts is **0** at every n : propagation alone settles these. That is worth stating plainly — these timings measure *encoding size*, not search difficulty, and should not be extrapolated to regimes where the solver has to work.

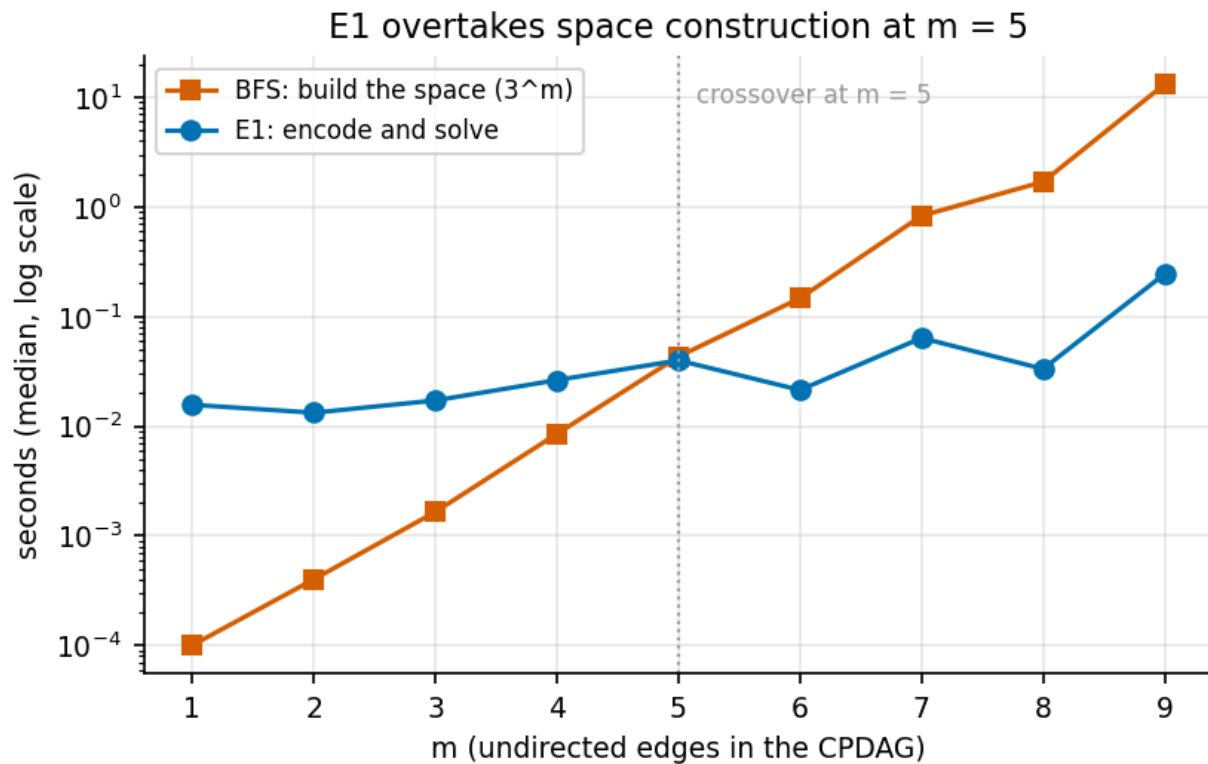


Build dominates; the SAT leg is negligible.

3.2 Against building the space: crossover at $m = 5$, then no contest

m	rows	space size	BFS build	E1 total	ratio
1	24	3	0.000 s	0.016 s	0.01×
2	84	6	0.000 s	0.013 s	0.03×
3	40	18	0.002 s	0.017 s	0.10×
4	62	30	0.009 s	0.026 s	0.32×
5	58	60	0.043 s	0.040 s	1.08×
6	30	98	0.147 s	0.021 s	6.88×
7	22	294	0.830 s	0.064 s	13.02×
8	22	210	1.696 s	0.033 s	51.00×
9	4	515	13.172 s	0.245 s	53.84×

Above $m = 9$ the space was not built at all (14 of 360 rows); 3^m closures is the binding cost, consistent with session 2's measured 3.97^k for BFS construction. This is the clean, uncontroversial win: anything that needs only a radius should not be building the space.



E1 overtakes space construction at $m = 5$.

3.3 The split: it is the radius, not k

A targeted high-k sweep, results/axisb3/highk.jsonl: 144 instances on dense Erdős-Rényi CPDAGs at $n = 8 \dots 18$ selected to have $|K(G_0)| \geq 10$. It reaches $|K(G_0)| = 51$, undirected components of **11 vertices** and densities up to **1.00**. local_up ran in a subprocess under a hard 60 s cap, so a timeout is recorded as a datum rather than silently dropped — omitting them would flatter the slower method. Where it completed it agreed with E1 on 117 of 117.

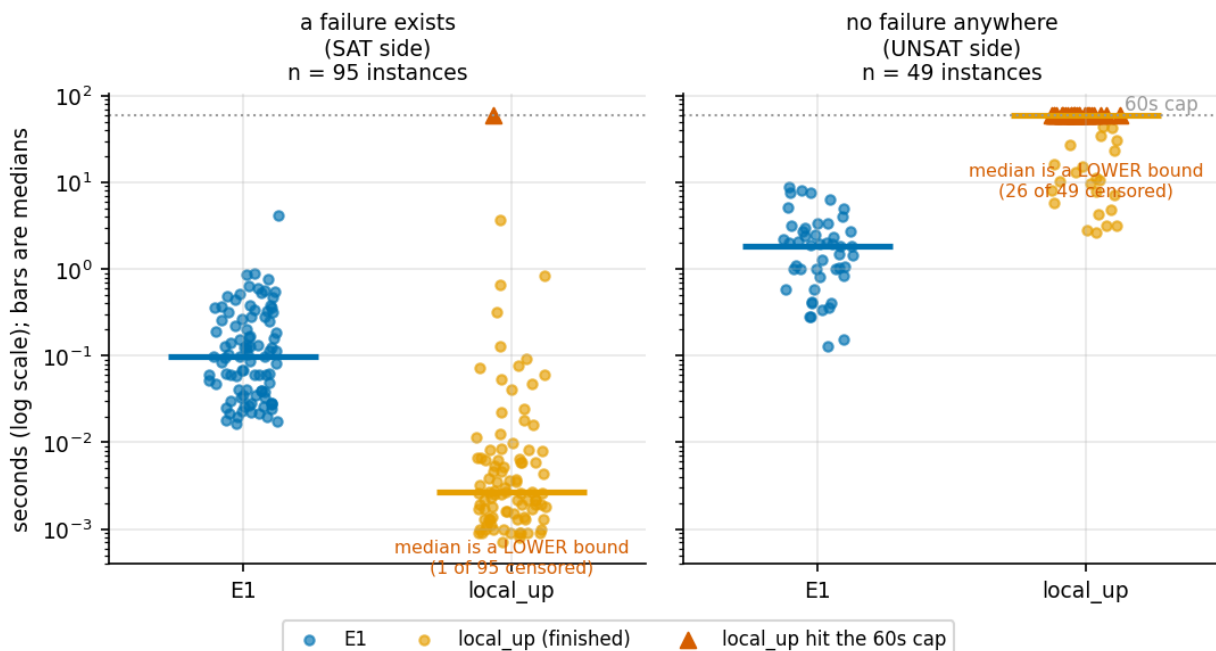
	rows	E1 median	E1 worst	local_up median	timeouts (>60 s)
a failure exists (SAT side)	95	0.099 s	4.117 s	0.0027 s	1 / 95
no failure anywhere (UNSAT side)	49	1.865 s	8.745 s	10.459 s (of 23)	26 / 49

That is the whole result in one table.

- **When a failure is near, local_up is unbeatable and E1 should not be used.** It stops at the first shell containing a failure; at radius 1 that is one expansion, while E1 must build an $O(n^4)$ -clause model before it can say anything. No encoding effort will close that gap.
- **When there is no failure, local_up must exhaust the entire up-set** and E1 wins by a margin not bounded above. On the 23 instances where local_up finished, the median speedup is **8.8x** and the maximum **117.9x**. On the other 26 it did not finish inside 60 s at all, while E1 answered every one — the worst, at $n = 16$ and $k = 33$, in 8.745 s.

So the earlier sessions' 1.59^k model for local_up describes its **worst case**, and the worst case is realised precisely on the degenerate instances. Its observed cost is dominated by how fast it stumbles onto a failure, which is why k alone does not predict the crossover and the radius does. This is also the practically useful half: finding a nearby failure is the easy question, and certifying that none exists is the one that supports a robustness claim.

The crossover is governed by the radius, not by k



The two regimes, separately. Censored runs are drawn at the cap and the local_up median on the right is therefore a lower bound.

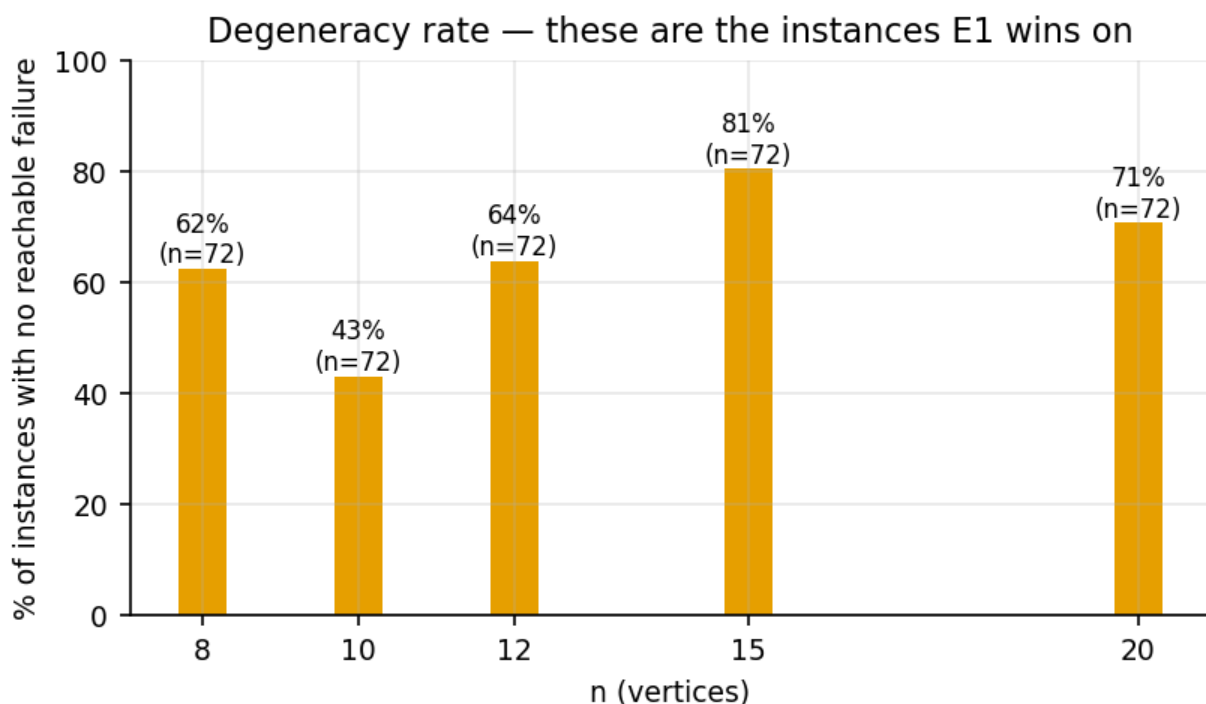
3.4 The ceiling, stated plainly

E1's cost is dominated by build time, which grows as $O(n^4)$ clauses in the vertex count and is essentially independent of k . Solve time is near zero on easy instances and becomes the bottleneck only on hard UNSAT instances at high k : median conflicts is 118,839 at $k = 33$, where the UNSAT leg is 8.48 s of a 8.745 s total.

- **$n \approx 20$ with $k \leq 12$** is comfortable: worst case 1.5 s.
- **$n = 18$ with k up to 51** is reachable, but the spread is wide: median 0.3 s with individual instances at 8.7 s.
- **Nothing here was pushed to failure.** No E1 run hit its time limit, so the ceiling above is where I stopped, not where the method breaks. I did not find E1's breaking point and should not imply a limit I did not measure.
- local_up's ceiling on the UNSAT side, by contrast, was measured: around $k = 12$ –15 at a 60 s budget.

3.5 Degeneracy

64.2% of the 360 general instances and 34.0% of the 144 high- k ones have no reachable failure at all. This is reported rather than filtered because it changes how every timing above should be read: these are the instances E1 wins on, and any aggregate that averages them together with the radius-1 cases is measuring a mixture of two very different problems. The UNREACHED sentinel is never averaged and never plotted as a number.



Degeneracy rate by n .

4. Conjecture 2, stress-tested where it could actually break

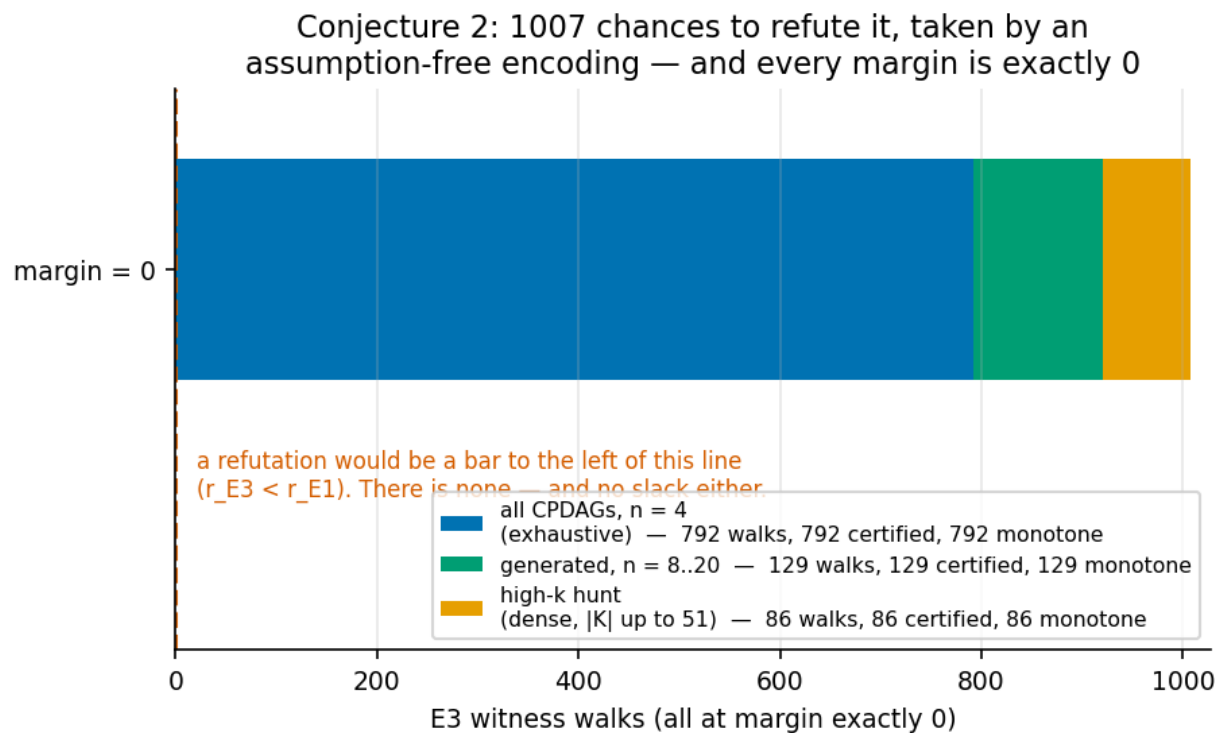
E3 assumes nothing. If it ever reached a failing state in fewer steps than any retraction-only search could, Conjecture 2 would be **refuted** — and with it the exactness of local_up and of E1. That was a live possibility going in, and it would have been the most important result the project has produced. **It did not happen.**

scope	E3 instances	certified	monotone	margins	counterexamples
all CPDAGs, $n = 4$ (exhaustive)	792	792	792	all 0	0
generated, $n = 8 \dots 20$	129	129	129	all 0	0
high-k hunt, dense	86	86	86	all 0	0

The hunt was deliberate, as the brief required: dense components, large components, high $|K(G_0)|$ — the regime session 2's Lemma-L slack statistics flagged as closest to the boundary. The largest instance on which E3 certified a walk had **$n = 18$, $|K(G_0)| = 51$, a 11-vertex undirected component at density 0.93**, and 2,125 variables, solved in 1.355 s. A directed hunt that finds nothing is much stronger evidence than an undirected one.

Three things must be said against over-reading this

- **Every margin is exactly zero.** Not a single instance had $r_{E3} > r_{E1}$ either. Conjecture 2 is never violated, but never violated *with room to spare* — it is tied everywhere. A conjecture that always ties is in a weaker evidential state than one with slack, and this is the opposite of the reassuring picture Lemma L gave in session 2, where only 9.0% of pairs were tight. Reported as a tie, not as a cushion.
- **E3's scope is bounded by radius, not by instance size.** E3 unrolls $k+1$ copies of the closure encoding, so it ran only where $r \leq 4$ (general sweep) or $r \leq 3$ (hunt). Instances at 5, 6 and 8 exist in the hunt and were **not covered**. The counterexample search is therefore a search over small-radius instances.
- **Agreement is not confirmation.** A shorter path through a hypothetical multi-orientation cover would be invisible to both E1 and E3. So this evidence bears on Conjecture 2 without touching **Lemma R**, which is what would rule such a cover out. Anti-Exchange Case B remains exactly as open as session 2 left it; this session did not try, and it was explicitly parked.



Every chance to refute Conjecture 2 that this session took, and the margin at each one.

What it does add. Session 2's evidence was ~2M comparisons inside enumerated spaces at $n \leq 5$. This adds instances an order of magnitude larger in every structural parameter, tested by a method that does not share the enumeration's assumptions and that *could* have refuted it. It also independently corroborates the monotone-path claim: 1,007 certified witness walks, every one monotone, on a search free to descend at every step.

5. What is safe to claim about tractability

Safe:

- The breakdown radius can be computed **without constructing the perturbation space**, and the answer agrees with brute force on every instance where brute force can still run (1,044 exhaustive at $n = 4$; 346 of 360 generated).
- For **certifying robustness** — proving no failure exists within the space — the encoding is the better method beyond roughly $k = 12$, where `local_up` stops finishing inside a 60 s budget and E1 continues to answer in seconds.
- An UNSAT answer at rung k is a **certificate that shells 0... k are clean**, produced without enumerating those shells.
- Radii from E3 are **assumption-free**; radii from E1 are conditional on Conjecture 2 and hence on Anti-Exchange Case B. The assumes field carries this, and it must not be dropped when a number is copied into a table.

Not safe, and not claimed:

- That the encoding is a general-purpose replacement for `local_up`. **It is not.** On the common case — a failure at radius 1 — it is roughly 37× slower and there is no prospect of closing that.
- Any statement about where E1 breaks down. It never hit a time limit here, so no breaking point was measured.
- That Conjecture 2 is confirmed. It survived a test that could have refuted it, at every margin exactly zero, over small-radius instances only.
- Any extrapolation of the timing tables to instances where the solver has to work. Median conflicts is 0 across the general sweep.

6. What went wrong

Reported at the same weight as the results, because in this session the bugs were the most instructive part: both encoding errors produced *plausible numbers*, neither crashed, and one erred in the direction that would have looked like a discovery.

- **The closure encoding was too strong.** Meek's rules as implications, with R3's and R4's undirectedness premises dropped, lost 67 of 133 spaces. Caught by comparing against the enumerated corrected space, not by inspection.
- **The collider clause was too weak, and it erred toward under-reporting robustness.** 144 of 26,304 cases. This makes radii too small, and a spuriously small E3 radius is exactly what a Conjecture 2 counterexample looks like. Caught only because Stage 2 was run as its own layer against a reference oracle, before any radius was computed.
- **UNREACHED conflated with “budget exhausted” in E3.** Found while writing the $n = 4$ comparison logic, not by a failing test.
- **A negated literal has no proto index on a clone.** Immediate crash in E2, trivially fixed — noted because it is the one bug here that announced itself.
- **A sed delimiter collided with a | in a type annotation**, and a later blanket replacement rewrote an import inside an embedded source string. Both caught at once by the linter, but both were edits made without reading the surrounding line first.

Not a bug, but worth recording: the first high- k sweep hung, because `local_up` on a degenerate instance at $k = 15$ must exhaust an up-set of up to 2^{15} states. I killed it and re-ran with `local_up` in a subprocess under a hard cap. The hang was not an obstacle to the measurement — **it was the measurement**, and turning it into a recorded timeout is what produced §3.3.

Pre-existing and untouched: 23 tests in `tests/test_adjustment.py`, `test_distances.py` and `test_radius.py` fail, and three more files fail to collect, all with `ImportError on TypeAlias / StrEnum`. These are the `REPO_INIT` scaffold stubs, which require Python 3.11; this machine runs 3.9.6.

Unrelated to this session. Everything else passes: 319 tests, including the 8 new ones in tests/sat/.

7. What this leaves for next time

- **A hybrid, and the data says exactly what it should be.** Run `local_up` with a small depth budget first; if it finds a failure it has the answer in microseconds. If it exhausts the budget without one, hand the instance to E1, which answers the UNREACHED question in seconds where `local_up` needs minutes or does not finish. The two are strong in disjoint regimes and the discriminator — has a failure been found yet — is available for free at runtime.
- **Reduce the build cost.** It dominates, and it is $O(n^4)$ in the vertex count while the answer depends only on the knowledge-intersected component. Restricting the encoding to that component before emitting clauses is the single highest-value change.
- **Extend E3's reach past small radii**, by incremental unrolling that reuses solver state across rungs rather than rebuilding.
- **The zero-margin observation deserves a look of its own.** Every $r_{E3} - r_{E1}$ is exactly 0 across 1,007 instances. That is consistent with Conjecture 2 being true, and it is also what one would see if the two encodings were, on these instances, searching the same object for a structural reason not yet identified. Worth understanding before the tie is presented as evidence.
- **Unchanged from session 2:** Anti-Exchange Case B is still the one open property and this session did not touch it. H4, the naive K-count baseline, is still the last unrun pre-registered hypothesis, now for a third session. Axis A remains paused.

8. Provenance

Solver: OR-Tools CP-SAT 9.15.6755, single worker, `random_seed=0`. Python 3.9.6. Machine timings are from one laptop and are reported alongside machine-independent counters wherever a claim rests on them. Every table in this document is computed from `results/axisb3/*.jsonl` at build time rather than transcribed, and `bkrobust.analysis.session3_verify` independently asserts that `report_axisb_sat.md` agrees with the same files.

Commits in this series

commit	date	subject
ad16807	2026-09-05	High-k hunt, session 3 figures, and <code>report_axisb_sat.md</code>
d9e283a	2026-09-05	Scaling study, and the encoding-correctness sections in <code>THEOREMS.md</code>
4d638cd	2026-09-05	E2 and E3, and the cross-encoding validation they exist for
8153913	2026-09-05	E1 SAT ladder: exact radii without building the space
215de2b	2026-09-05	Whitelist <code>results/axisb3</code> BEFORE writing any results
68c395e	2026-09-05	Propagate the updated Case B scope (36,094) to every deliverable
60d63eb	2026-09-05	Fix the PDF to match the report: density gap closed, not open
afa7400	2026-09-05	Close the $n=5$ density gap: 135 CPDAGs by sweep, the last by the proof chain
e984e18	2026-09-05	Correct the $n=5$ figure: the corrected sweep completed at 1,977,820 comparisons
940da08	2026-09-05	Session 2 deliverables: report, PDF, figures, <code>NEXT.md</code> , scaling and bounds results
0a4fa92	2026-09-05	Track <code>results/axisb2</code> (recurrence of last session's gitignore bug)
b92ebd5	2026-09-05	Add the Axis B scaling harness (two accountings, machine-independent counters)
7be840c	2026-09-04	Conjecture 2 reduced to a single local property; Property S and Lemma R proved
79d7367	2026-09-04	<code>THEOREMS.md</code> : Conjecture 2 reduces to upper semimodularity; join formula proved
748f2e1	2026-09-04	Task 0: the space excluded legitimate knowledge states; corrected builder added
007e6dd	2026-09-04	Add <code>SESSION_REPORT.pdf</code> : exhaustive 23-page record of the Axis A / Axis B session
3a61b40	2026-09-03	Frontier sweep complete: O^* never beaten across 249,732 exhaustive instances
d4e0c14	2026-09-03	Session deliverables: report, <code>NEXT.md</code> , H6 calibration, PLAN revisions
72d8bff	2026-09-03	Correct the H2 statistic: the frontier is non-flat, but O^* is never beaten

commit	date	subject
ca44152	2026-09-03	Axis A results: H1 saturation ~80%, H2 frontier flat in 0 of 134,140 instances
019c610	2026-09-03	Axis A first run: partial results, two gate bugs found and sent back
1bb3ac8	2026-09-03	Record: Meek closure can fall outside enumerate_space in 0.09% of cases
4cf4020	2026-09-03	Conjecture 2: no counterexample in 1.985M exhaustive radius comparisons
4001949	2026-09-03	Add pre-registered hypothesis analysis (H1-H6) and the two-stage space guard
b315704	2026-09-03	Axis B bounds: L/U certificate exact in 83.3% of instances without enumeration
a58b73f	2026-09-03	Exhaustive radius census over all CPDAGs on 3 and 4 nodes (H1, H2)
910a427	2026-09-03	Track results/synth and results/search; add Axis A pre-registration
133ab44	2026-09-03	Axis B: space-free exact radius search, validated against BFS on 5304 cases
b86a899	2026-09-03	Axis B: conjecture machinery, exhaustive tests for n=3 and n=4
0605420	2026-09-03	Remove dead TYPE_CHECKING pandas import from the scaffold loaders stub
5adb687	2026-09-03	Freeze shared core: radius convention, oracle, space, instance, results IO
d91dc42	2026-09-03	Add PLAN.md and LOG.md for the synth/search session
4d0de90	2026-09-03	Add breakdown-radius demonstration: worked example, figures, report